

CONTACT	Email: sunayj@sas.upenn.edu Website: sunayjoshi.github.io GitHub: github.com/sunayjoshi Google Scholar: scholar.google.com/citations?user=8fTMwcUAAAAJ
RESEARCH INTERESTS	Statistical machine learning; graphical and network models; high-dimensional asymptotic statistics; statistical uncertainty quantification; applied selective inference; foundations of LLM post-training; applied stochastic optimal control; applications to financial econometrics and macroeconometrics.
EDUCATION	University of Pennsylvania , Ph.D. in Applied Mathematics 2024–2028 (expected) - Advisors: Edgar Dobriban and Hamed Hassani - Committee: Edgar Dobriban, Hamed Hassani, Zhimei Ren, Aaron Roth - Title: “Conformal and Selective Inference for Reliable AI” Princeton University , B.A. in Mathematics with Highest Honors (<i>summa cum laude</i>) 2020–2024 - Minors: Applied and Computational Mathematics; Applications of Computing - Senior Thesis Advisor: Allan Sly - Title: “Alignment of Correlated Random Graphs” - GPA: 3.98/4.00
PUBLICATIONS AND PREPRINTS	Two working papers on (1) statistical proof verification via multiple hypothesis testing; (2) finite-sample theory of synthetic data for estimating linear models. (Both intended for submission to AISTATS 2027.) Sunay Joshi, Ilia Popov, Shuyi Wang, Edgar Dobriban, Hamed Hassani, and Zhimei Ren. “Conformal Factuality with Learned Structure.” Submitted to <i>ICLR</i>, 2027. Sunay Joshi, Tao Wang, Hamed Hassani, and Edgar Dobriban. “Risk-Controlled Post-Processing of Decision Policies.” Submitted to <i>NeurIPS</i>, 2026. arXiv:2605.06479. Sunay Joshi, Yan Sun, Hamed Hassani, and Edgar Dobriban. “MultiRisk: Multiple Risk Control via Iterative Score Thresholding.” Revise and resubmit at <i>JASA</i>. arXiv:2512.24587, 2025. Georgy Noarov, Soham Mallick, Tao Wang, Sunay Joshi, Yan Sun, Yangxinyu Xie, Mengxin Yu, and Edgar Dobriban. “Foundations of Top-k Decoding for Language Models.” <i>NeurIPS</i>, 2025. arXiv:2505.19371. Sunay Joshi, Shayan Kiyani, George J. Pappas, Edgar Dobriban, and Hamed Hassani. “Conformal Inference under High-Dimensional Covariate Shifts via Likelihood-Ratio Regularization.” <i>NeurIPS</i>, 2025. arXiv:2502.13030. Gary Hu, Julian Gold, Uthsav Chitra, Sunay Joshi, and Benjamin J. Raphael. “Anomaly Detection in Spatial Transcriptomics via Spatially Localized Density Comparison.” <i>Bioinformatics</i>, 41(Supplement 1):i493–i501, 2025. doi:10.1093/bioinformatics/btaf242. Marissa Gee, Nicolas Gonzalez-Granda, Sunay Joshi, Nagaprasad Rudrapatna, Anne Somalwar, Stephen P. Ellner, and Alexander Vladimirsky. “Navigating the Landscape of Fear.” <i>bioRxiv</i>, 2024. doi:10.1101/2024.08.15.608007.
TALKS AND PRESENTATIONS	Poster Presentation, STAI-X Conference, Harvard University August 2026 “Risk-Controlled Post-Processing of Decision Policies.” Poster Presentation, NeurIPS Conference, San Diego December 2025 “Conformal Inference under High-Dimensional Covariate Shifts via Likelihood-Ratio Regularization.” Virtual Talk, EnCORE Collaboration Workshop, UC San Diego September 2025 “Conformal Inference under High-Dimensional Covariate Shifts via Likelihood-Ratio Regularization.” Group Talk, Dobriban Stat-ML Reading Group, University of Pennsylvania January 2026 “Introduction to LLM reasoning.”

TEACHING

- **Undergraduate Course Assistant**, Princeton University September 2023–December 2023
 - Served as one of two undergraduate assistants for MAT 340: Applied Algebra.
 - Led weekly problem sessions on introductory group theory and representation theory.

REVIEWING

- Reviewer: *Journal of the American Statistical Association* (JASA), 2025; STAI-X 2026; NeurIPS 2026; OPT Workshop at NeurIPS 2026; ICML 2026 Workshop on AI4Research.

SERVICE AND OUTREACH

- **Volunteer**, ICML 2026 Workshop on AI4Research 2026
- **Mentor**, Directed Reading Program (DRP) Spring 2026, Fall 2026
 - Topics: high-dimensional statistics, distribution-free statistics.
- **Organizer**, Dobriban Stat-ML Reading Group September 2024–April 2025
 - Organized logistics for Prof. Edgar Dobriban’s statistics and machine learning reading group.
 - Selected speakers for weekly meetings, managed shared materials, and made announcements.
- **Volunteer**, University of Pennsylvania Math Circles September 2024
 - Provided after-school mathematics enrichment activities for students at West Philadelphia High School.
- **Advising Chair**, Princeton Mathematics Club April 2023–May 2024
 - Organized course advising sessions for undergraduates in the mathematics department.
 - Organized the Mentoring Möbius program connecting undergraduates with mathematics graduate students.
- **Problem Czar**, Princeton Math Contest 2022
 - Served as lead problem writer for Princeton’s annual high school mathematics contest (PUMaC).
 - Held regular problem-writing meetings, selected final contest problems, and wrote 48 problems over four years.

AWARDS AND HONORS

- Sigma Xi, 2024: elected to national science honor society.
- Phi Beta Kappa, 2024: elected to the oldest academic honor society in the United States.
- NSF Graduate Research Fellowship Program Honorable Mention, 2024.
- Andrew H. Brown Prize, 2023: awarded by the Princeton Mathematics Department to outstanding juniors in mathematics.
- Class of 1861 Prize, 2022: awarded by Princeton to the sophomore with the best record on the Putnam Examination.
- Shapiro Prize for Academic Excellence, 2021: recognizes outstanding academic achievement by Princeton undergraduates in their first or second years of study.
- Rank 150, William Lowell Putnam Mathematical Competition, 2021.
- Silver Medalist, USA Physics Olympiad, 2019.
- Qualifier, USA Junior Math Olympiad, 2018.

TECHNICAL SKILLS

- *Programming*: Python, Java, C++, Linux, Bash, GitHub, agentic programming workflows.
- *Typesetting*: LaTeX.
- *Languages*: English (native), Spanish (intermediate), Hindi (beginner).